MA1513 Weekly Self Practice Set 4 Answer

1. (a) Given that $T : \mathbb{R}^3 \to \mathbb{R}^3$ is a linear transformation such that

$$T\left(\begin{pmatrix}1\\0\\0\end{pmatrix}\right) = \begin{pmatrix}2\\3\\1\end{pmatrix}, \quad T\left(\begin{pmatrix}0\\1\\0\end{pmatrix}\right) = \begin{pmatrix}0\\2\\1\end{pmatrix}, \quad T\left(\begin{pmatrix}0\\1\\1\end{pmatrix}\right) = \begin{pmatrix}0\\2\\2\end{pmatrix}.$$

Write down the standard matrix $\boldsymbol{A}$ of T. Briefly explain how you find $\boldsymbol{A}$.

Answer

From the given information, we have

$$\boldsymbol{A}\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}3\\2\\1\end{pmatrix}, \quad \boldsymbol{A}\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\3\\1\end{pmatrix},$$

$$\boldsymbol{A}\begin{pmatrix}0\\0\\1\end{pmatrix} = \boldsymbol{A}\begin{pmatrix}0\\1\\1\end{pmatrix} - \boldsymbol{A}\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\2\\2\end{pmatrix} - \begin{pmatrix}0\\2\\1\end{pmatrix} = \begin{pmatrix}0\\0\\1\end{pmatrix}.$$

By stacking, we get $\boldsymbol{A} = \boldsymbol{A}\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix} = \begin{pmatrix}2&0&0\\3&2&0\\1&1&1\end{pmatrix}$.

(b) Without solving characteristic polynomial or using MATLAB, find all the eigenvalues of the standard matrix $\boldsymbol{A}$ in part (a).

Answer

Since the matrix $\boldsymbol{A} = \begin{pmatrix}2&0&0\\3&2&0\\1&1&1\end{pmatrix}$ is lower triangular, the eigenvalues are the diagonal entries $2, 1$.

(c) Find a basis for each of the eigenspaces of $\boldsymbol{A}$ associated with all the eigenvalues (without using MATLAB). Show your working.

Answer

First eigenvalue:

For $\lambda = 1$, solve the linear system $(1\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x} = \boldsymbol{0}$,

$$\begin{pmatrix}-1&0&0\\-3&-1&0\\-1&-1&0\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix} \Leftrightarrow \left(\begin{array}{ccc|c}-1&0&0&0\\-3&-1&0&0\\-1&-1&0&0\end{array}\right) \Leftrightarrow \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}0\\0\\t\end{pmatrix} \quad t \in \mathbb{R}$$

Hence $E_1 = \text{span}\left\{\begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$ and a basis for E_1 is $\left\{\begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$.

Second eigenvalue:

For $\lambda = 2$, solve the linear system $(2\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x} = \boldsymbol{0}$,

$$\begin{pmatrix} 0 & 0 & 0 \\ -3 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ -3 & 0 & 0 & 0 \\ -1 & -1 & 1 & 0 \end{array} \right) \Leftrightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ t \\ t \end{pmatrix} \quad t \in \mathbb{R}$$

Hence $E_2 = \text{span}\left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$ and a basis for E_2 is $\left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$.

(d) Using MATLAB with <u>format short</u> option for this part.

(i) Type the command $\gg$ `[P1 D1]=eig(A)`. (You will get a matrix $P1$ where the columns are eigenvectors; and a diagonal matrix $D1$ where the diagonal entries are eigenvalues.) Write down the two columns of $P1$ that give the eigenvectors corresponding to the larger eigenvalue of $\boldsymbol{A}$.

(ii) Type the command $\gg$ `[P2 D2]=eig(sym(A))`. Write down the column of $P2$ that gives the eigenvector corresponding to the larger eigenvalue of $\boldsymbol{A}$.

(iii) Explain the differences that you have observed in (i) and (ii).

Answer

(i) The two columns for eigenvalue 2 are $\begin{pmatrix} 0 \\ 0.7071 \\ 0.7071 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ -0.7071 \\ -0.7071 \end{pmatrix}$.

(ii) The column for eigenvalue 2 is $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

(iii) Using command `[P1 D1]=eig(A)`, it will always give a square matrix $P1$, regardless of whether the columns are linearly independent or not. Using command `[P2 D2]=eig(sym(A))`, it will give $P2$ with only linearly independent columns, which may not be a square matrix. Note that, the two columns in $P1$ are both scalar multiples of the column in $P2$. They are linearly dependent eigenvectors in the eigenspace E_2.